🩺 Trust-First Triage Desk

Verify what **10,088** Indian healthcare facilities actually claim — with evidence and **honest uncertainty**

Databricks × Virtue Foundation Hackathon 2026 · Track 1: Facility Trust Desk

Live app: trust-first-triage-desk-108684035875991.aws.databricksapps.com

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The problem

- The dataset gives **10,088 facility records** scraped from the open web (Bright Data → GenAI extraction → entity resolution).
- The `capability`, `equipment`, and `procedure` fields are **claims, not verified facts** — coverage ranges 100% → 25%.
- A hospital can list "ICU" in free text while its own description says "*ICU cases referred elsewhere.*"

> **143M** people await surgery in LMICs each year · **2.88B** DALYs lost to inadequate care.

VF Match already shows *where the deserts are. The unanswered question: can this facility actually do what it claims?*

What it does

A **Databricks App for a non-technical planner**:

- **Triage view** — pick one of **12 capabilities** (ICU, NICU, maternity, emergency, oncology, trauma + 6 more), filter by state/city. Facilities ranked by trust and color-coded ✅ **Verified** · ⚠️ **Unclear** · ❌ **Contradicted**.
- **Facility Detail** — exact claim text + every supporting/contradicting snippet from the facility's *own* fields, with a trust gauge.
- **Operations & access** — 24/7, wheelchair, ambulance, blood bank… and flags **closed / under-construction** before a planner commits.
- **Persistent work** — verify / reject / notes saved to **Lakebase Postgres**.

Every badge cites a quoted snippet. We never present weak evidence as fact.

Data foundation — messy source → trustworthy silver

Bronze → Silver medallion, idempotent and re-runnable (`notebooks/02_build_silver.py`):

- **Parse** the semi-structured JSON-array strings (`capability` , `equipment` , `procedure` , `specialties`) → `ARRAY<STRING>` with `FROM_JSON` ; `TRY_CAST` numerics.
- **Geo-validate** every lat/lng against an India bounding box (6–37° N, 68–98° E) → `has_valid_coords` . Caught a "Kerala" hospital geocoded to the **North Atlantic Ocean**.
- **Resolve state honestly** — `address_stateOrRegion` sometimes holds a *district*. Pincode lookup overrides it; `state_source ∈ {pincode, source, unresolved}` records provenance.
- **Dedup India Post** — 165,627 post offices → **19,586 pincodes** (Head Office > Post Office > Branch Office).
- **NFHS-5 cleaning** — `*` → NULL (suppressed), `(29.5)` → low-sample flag.

Provenance columns (`state_source` , `has_valid_coords`) make every cleaned value auditable — the honesty tax that earns trust.

How it works — claim → evidence → trust

```
Source arrays              Three-tier classifier        Evidence scan             Trust score
(capability,       →       1. parse JSON arrays    →    match capability   →      conf, supports,
 specialties,              2. rules + codes              terms in the             contradicts
 equipment,                3. LLM fallback               facility's OWN     →     → Verified /
 description)               (Llama 3.3 70B)              text, ± window             Unclear /
                                                          polarity                  Contradicted
```

```
trust=max(0,min(1,(conf-0.20)+min(0.30,0.08s)-min(0.80,0.30c)))
```

Base starts *below* extraction confidence — one mention isn't proof. Verified needs trust ≥ 0.75 **and** ≥ 2 corroborations. Any contradiction with s ≤ c → Contradicted.

Honest uncertainty is the point

Status	What it means	Example
✅ **Verified**	quantified + corroborated	*"22-bed Level II ICU with 11 ventilators"*
⚠️ **Unclear**	claimed via code only, no prose backing	specialty code `criticalCareMedicine`, nothing else
❌ **Contradicted**	denied or not operational	*"NICU facility not available"* · *"trauma centre under construction"*

- Tightening thresholds moved **3,000+ facilities from "verified" to "unclear"** — and that's the **right answer**.
- The contradiction detector scans **both before and after** the mention → catches "*ICU cases referred to NMC Hospital.*"

Operations & access — what a planner needs next

Beyond clinical capability, surfaced live from the facility's own text (each **cited**):

- 🕒 Open 24/7 · ♿ Wheelchair / disabled access · 🚑 Ambulance
- 🩸 Blood bank · 💊 Pharmacy / Jan Aushadhi · 💳 Cashless / Ayushman
- 🚧 **Under construction** · ⛔ **Temporarily closed** (shown as warnings)

Sidebar filter — "Must offer" + "Hide closed / under-construction". Filters on *stated evidence* — absence is never treated as denial.

Evidence it works

20 hand-labeled scenarios from real Marketplace rows — reproducible, offline, deterministic (`python -m eval.run_eval`):

Metric	Result
Overall accuracy	**17 / 20 = 85%**
Contradiction **precision**	**100%**
Contradiction **recall**	**100%**

- A planner is **never told a denied/under-construction service exists**, and **never misses one** the text states.
- The 3 remaining errors are all in the **safe direction** (under-claiming, never over-claiming).

Built on Databricks Free Edition — end to end

- **Medallion Delta** (Bronze → Silver → Gold) on Unity Catalog
- **Foundation Model API** (Llama 3.3 70B) for the LLM classification tier — no external keys
- **Lakebase Postgres** (+ pgvector) for persistent planner verifications, notes, shortlists
- **Databricks Apps + Streamlit** — live URL within 2 hours; push → sync → deploy
- **Provenance baked into the schema:** `state_source`, `extraction_method`, `has_valid_coords` — *how* each value was derived

All on Free Edition — no external keys or paid tiers. See the "Data foundation" slide for the silver-layer cleaning that makes every score auditable.

Impact & what's next

- **Complements VF Match:** discovery (*where*) + verification (*whether*) in one workflow.
- **District context** (NFHS-5): Jhabua, MP at **38.6%** women's literacy — find *verified* coverage where burden is highest.
- **Unlocks the other tracks:** verified `gold_facility_trust` makes a Medical Desert Planner and Referral Copilot trustworthy by side effect.

Roadmap: PICU/ICU disambiguation · description-supported boolean · Mosaic AI agent for NL queries · daily Lakeflow refresh with flip-to-contradicted alerts.

Verify what facilities actually claim. Trust, with the receipts.